

1. Project Title

Carbon Footprint Awareness and Action Guide

2. Project Overview

The Carbon Footprint Awareness and Action Guide is an AI-powered dashboard developed using the AWS PartyRock platform and Whiskers AI. The application helps users understand the environmental impact of their daily activities, estimate carbon emissions, and receive personalized recommendations to reduce their carbon footprint.

Users can enter details about activities related to transportation, energy usage, food consumption, or waste generation along with their weekly frequency. The system then analyzes the information and provides actionable sustainability insights.

3. Problem Statement

Many individuals are unaware of how their daily habits contribute to carbon emissions. This lack of awareness makes it difficult to adopt environmentally responsible practices.

The objective of this project is to:

- Increase awareness about carbon emissions.
 - Help users track carbon-generating activities.
 - Provide personalized recommendations for reducing environmental impact.
 - Encourage sustainable lifestyle choices.
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4. Technology Stack

Component	Technology Used
Development Platform	AWS PartyRock
AI Assistant	Whiskers AI
User Interface	PartyRock Widgets
Analysis Engine	Generative AI
Deployment	AWS PartyRock Cloud Platform

5. Application Features

User Inputs

- Activity Category
- Activity Details
- Frequency per Week

AI-Generated Outputs

- Carbon Impact Analysis
- Personalized Reduction Tips
- Alternative Eco-Friendly Solutions
- Carbon Footprint Assistant Guidance

6. System Workflow

1. User selects an activity category.
2. User enters activity details.
3. User specifies frequency per week.
4. Whiskers AI analyzes the activity.
5. Carbon emission impact is estimated.
6. Personalized recommendations are generated.
7. Alternative sustainable solutions are suggested.

7. Dashboard Components

Input Widgets

- Activity Category Dropdown
- Activity Details Text Box
- Frequency per Week Slider

Output Widgets

- Carbon Impact Analysis
- Personalized Reduction Tips
- Alternative Solutions
- Carbon Footprint Assistant

8. Evaluation Parameters

Code Quality

- Clean widget organization and logical workflow.
- Proper naming conventions for all components.
- Easy-to-understand user interface.

Security

- No collection of sensitive personal information.
- User data is processed only for generating recommendations.
- AWS PartyRock's secure cloud environment is utilized.

Efficiency

- Instant AI-generated analysis.
- Lightweight dashboard design.
- Minimal user input requirements.

Testing

Test Case	Input	Expected Result	Status
Transportation Activity	Drive car 20 km/day	Carbon impact analysis generated	Passed
Energy Usage	AC usage 6 hours/day	Energy-saving recommendations	Passed
Food Consumption	Meat consumption 5 times/week	Sustainable food suggestions	Passed
Waste Generation	Plastic usage daily	Waste reduction tips	Passed

Accessibility

- Simple and intuitive interface.
- Easy-to-read layout.
- Minimal technical knowledge required.
- Suitable for users of different age groups.

Problem Statement Alignment

The application successfully addresses the challenge by:

- Educating users about carbon emissions.
 - Tracking environmentally impactful activities.
 - Providing personalized recommendations.
 - Encouraging sustainable decision-making.
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9. Sample Use Case

User Input

- Category: Energy Usage
- Activity Details: AC used for 8 hours daily
- Frequency: 7 days per week

AI Output

- High energy consumption detected.
 - Estimated increase in carbon emissions.
 - Suggestions:
 - Increase AC temperature by 1–2°C.
 - Use energy-efficient appliances.
 - Reduce unnecessary AC usage.
 - Consider renewable energy sources.
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10. Results and Benefits

Benefits

- Promotes environmental awareness.
- Encourages eco-friendly habits.
- Provides personalized sustainability insights.
- Helps users make informed decisions.

Outcome

The Carbon Footprint Awareness and Action Guide effectively leverages AWS PartyRock and Whiskers AI to deliver an interactive, user-friendly platform that supports carbon footprint reduction and environmental sustainability.

11. Conclusion

This project demonstrates how Generative AI can be used to address environmental challenges. By combining AWS PartyRock and Whiskers AI, the application provides real-time carbon footprint analysis and practical recommendations, empowering users to make sustainable choices and contribute to a greener future. Deployment Sample snapshot

Carbon Footprint Awareness and Action Guide

Welcome

Welcome to the Carbon Footprint Awareness Platform. Describe an activity or habit you engage in, and our AI will analyze its environmental impact, estimate CO2 emissions, and provide personalized practical tips to help you make greener choices -- one step at a time.

Activity Category

Select a category...

Activity Details

Describe your specific activity or habit, e.g. drive 30 miles daily, eat meat 5 times a week, use AC all day.

Frequency per Week

1

Carbon Footprint Analysis

Fill in **Activity Category**, **Activity Details**, and **Frequency per Week** to generate content.

Personalized Reduction Tips

Fill in **Activity Category**, **Activity Details**, and **Frequency per Week** to generate content.

Alternative Solutions

Fill in **Activity Category**, **Activity Details**, and **Frequency per Week** to generate content.

Carbon Footprint Analysis

Fill in **Activity Category**, **Activity Details**, and **Frequency per Week** to get started.

Carbon Footprint Awareness and Action Guide

Welcome

Welcome to the Carbon Footprint Awareness Platform! Describe an activity or habit you engage in, and our AI will analyze its environmental impact, estimate CO2 emissions, and provide personalized, practical tips to help you make greener choices — one step at a time.

Activity Category: Waste Management

Activity Details: Food Waste, Recycling, Composting, E-Waste, Paper Waste, Garbage Disposal, Reusable Items Usage

Frequency per Week: 0 to 7

Carbon Impact Analysis

Environmental Impact Analysis: Waste Management Practices

Your Activity Profile

Category: Waste Management | Frequency: 1 time per week across multiple waste streams

1. Estimated CO₂ Equivalent Emissions

Category	Emissions
E-Waste (improper handling risk)	+5-15 kg CO ₂ e/year
Garbage Disposal (general)	+30-40 kg CO ₂ e/year

Personalized Reduction Tips

Your Waste Management Sustainability Tips

Great news — the fact that you're already engaging with waste management practices **once a week** is a fantastic starting point! Even small, consistent changes in this area can make a meaningful difference. Here are your personalized tips to level up your impact:

Tip 1: "The Sorting Superpower"

What to do: Set up a simple 3-bin system at home for plastics, paper, and food waste. Spend just 5 minutes during your weekly routine making sure items are correctly sorted before disposal.

Impact: Proper sorting can divert up to **1,000 lbs of waste per household per year** from landfills, reducing methane emissions significantly.

Difficulty: Easy

Tip 2: "Compost Your Carbon"

What to do: Start a small countertop compost bin for food scraps like peels, coffee grounds, and eggshells. Drop it off at a local composting facility or use it in a garden weekly.

Impact: Composting food waste instead of landfilling it can **reduce CO₂-equivalent emissions by up to 0.5 kg**.

Alternative Solutions

Your Eco-Friendly Waste Management Guide

Based on your current habits across **8 waste-related activities** at a frequency of **1 time per week**, here's how to build momentum toward a more sustainable lifestyle. Every small shift adds up — let's make it easy!

Immediate Swaps

Changes you can start today with minimal cost or effort

Plastic Waste

- Switch to reusable bags, bottles, and containers for your weekly shopping or errands. Single-use plastics are among the top pollutants in oceans and landfills — even reducing by one item per outing makes a measurable difference.
- Choose products with minimal or recyclable packaging when shopping. Look for the recycling symbol and favor glass, cardboard, or aluminum over plastic.

Food Waste

- Plan one weekly meal prep session to use ingredients before they spoil. Food waste in landfills produces methane, a potent greenhouse gas — using what you buy is one of the highest-impact habits you can build.

Carbon Footprint Assistant

Hello! I am your Carbon Footprint Assistant, here to help you on your sustainability journey. I have got the details about your activity and I am ready to dive deeper into any questions you have — whether it is understanding the science behind emissions, overcoming barriers to change, exploring other areas of your lifestyle, or anything else climate-related. What would you like to explore?

Deployment Link:

https://partyrock.aws/u/kgselvamtech/S_ZwobM4I/Carbon-Footprint-Awareness-and-Action-Guide

Github Repository Link:

<https://github.com/krithikgokuls/Carbon-Footprint-Awareness-and-Action-Guid>